

Listed below are the top open source tools for monitoring Kubernetes that businesses rely on.

Prometheus

This is the most popular open source database among users of Kubernetes. What sets Prometheus apart from other databases is its multi-dimensional data mode, alert mechanisms and high-end querying language. It is a part of the Cloud Native Computing Foundation and is one of the most popular tools for Kubernetes monitoring. It offers a variety of client libraries supporting over ten languages. Its existing exporters allow bridging of third party data into Prometheus with systems like HAProxy and Docker.

Benefits:

- Supports the discovery and monitoring of scheduled services
- Multi-dimensional data model
- Supports both black box and white box monitoring

Grafana

This tool helps to break down the Kubernetes metrics and create dashboards. It is a very robust tool for Kubernetes. Grafana is usually set up along with Graphite or InfluxDB.

Benefits:

- Robust
- Has filtering capabilities
- Data-specific querying
- Visualisation through dashboarding
- Simple to set up

Kubelet

Kubelet bridges the gap between the master and the nodes in a Kubernetes cluster. It is a node agent that runs on every node while maintaining a set of pods. Kubelet communicates with the APIs to get statistics on resource utilisation, events and pods' status.

Benefits:

- Fetches the information about each container's usage through cAdvisor

- Ensures that the performance of the container matches the specifications in the PodSpecs

Kube-ops-view

Although Kube-ops-view is not a monitoring tool, it can give a clear picture of the operations of the Kubernetes clusters. It shows the various nodes running, their resources and capacity. It also shows their status and the pods deployed on those nodes.

Benefits:

- Easy to deploy and use
- Gives a complete picture of multiple Kubernetes clusters and operations
- Gives the status of pods and nodes

Heapster

This is one of the most used tools for monitoring Kubernetes clusters. It generates metrics about various Kubernetes components and can also integrate with other tools such as Prometheus.

Benefits:

- Supports various data sinks to show the collected metrics
- The code of the data sinks resides within its own repository

Weave Scope

Weave Scope works as a monitoring and troubleshooting tool for Kubernetes. It shows a map depicting containers, processes and the hosts of a Kubernetes cluster. It has a nice user interface where you can enter diagnostic commands. It is a good tool for those in search of in-depth details of the infrastructure, connections and components of a data system.

Benefits:

- Zero configuration and interactive user interface
- Gives a visual representation of the Kubernetes cluster
- Shows the infrastructure and connections among the various components

cAdvisor

cAdvisor or Content Resource Advisor is a performance analysis tool for various resources. It collects data and statistics about various machines, and is deployed at the node level. It is basic and hence ideal for beginner level usage, but is limited when it comes to in-depth functionality. It is installed on all cluster nodes for collecting, processing and exporting usage metrics about CPU, memory and bandwidth usage at the node level, on running containers.

Benefits:

- Easy to deploy and use
- Built inside Kubernetes

kubewatch

True to its name, kubewatch looks out for Kubernetes events and notifies them to all the endpoints. It will note any changes made to the Kubernetes resources.

Benefits:

- Easy to deploy
- Provides support for several endpoints

kube-state-metrics

The kube-state-metrics tool listens to the API of Kubernetes, and generates metrics about the status of pods, nodes and so on. It exports the metrics on an HTTP endpoint, and is easy to integrate with Prometheus due to this simple functionality. It is simple to use but can only be used for generating metrics in a limited scope.

Benefits:

- Easy to use
- Integrates well with Prometheus

Jaeger

Jaeger is an ideal tool to manage problems in complex distributed systems. It can help monitor and troubleshoot processes easily. The problems addressed by this tool are the optimisation of latency, transaction monitoring, and many more.